



VMWARE VSPHERE 7.0: INSTALL, CONFIGURE, MANAGE



Lab 05: Adding and Managing Virtual Hardware

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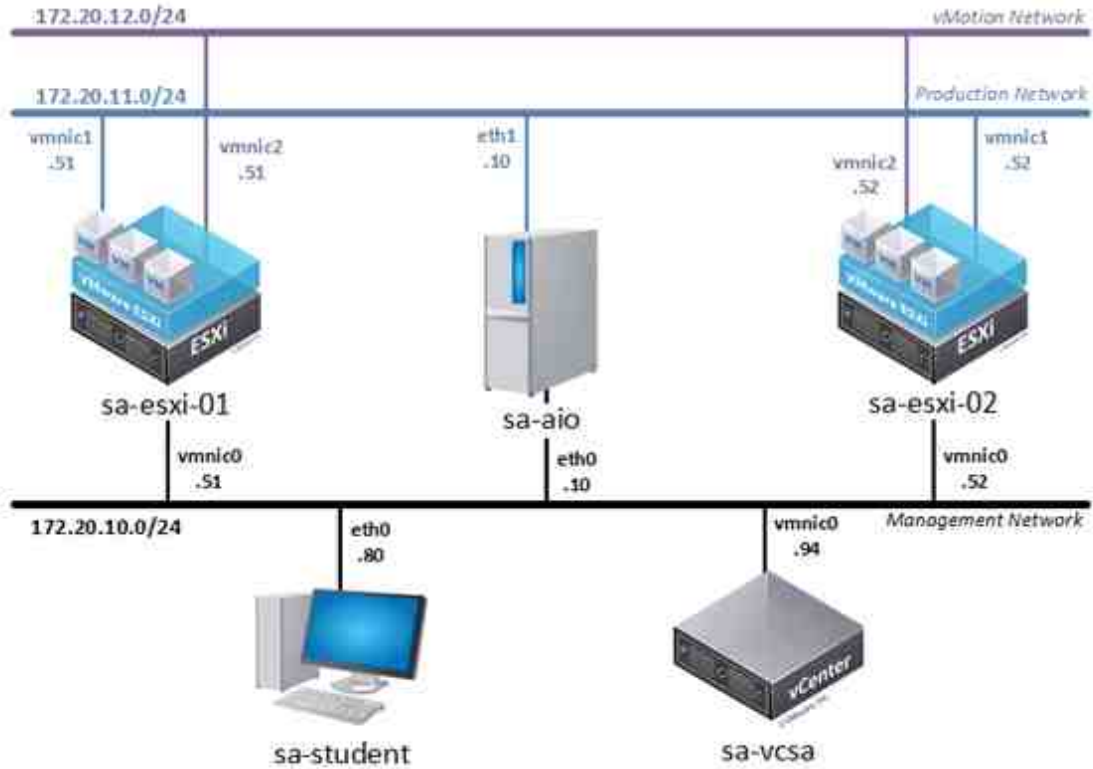
Introduction

In this lab, you will utilize the VMware Host Client to examine a virtual machine's configuration, add virtual hardware to a virtual machine, and delete virtual hardware from a virtual machine.

Objectives

-) Examine a Virtual Machine's Configuration
-) Create and add Virtual Hard Disks to a Virtual Machine
-) Differentiate a Thin-Provisioned and Thick-Provisioned Disks
-) Delete virtual hardware

Lab Topology



Lab Settings

The information in the table below will be needed in order to complete the lab. The task sections below provide details on the use of this information.

Virtual Machine	IP Address	Account	Password
sa-esxi-01	eth0: 172.20.10.51	root	Train1ng\$
sa-esxi-02	eth0: 172.20.10.52	root	Train1ng\$
sa-aio	eth0: 172.20.10.10	sysadmin	vmware123
sa-student	eth0: 172.20.10.80	sysadmin	vmware123
sa-vcasa	eth0: 172.20.10.94	sysadmin@vclass.local	vmware123

1 Examine a Virtual Machine's Configuration

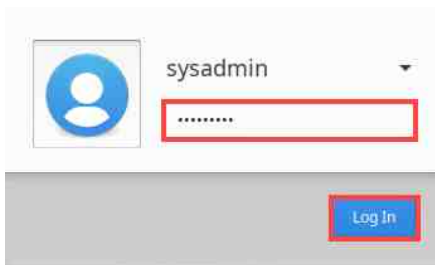
In this task, you will use the VMware Host Client to examine a VM's configuration. Viewing a VM's configuration is useful for general VM maintenance and troubleshooting purposes.

1. Launch the sa-student virtual machine to access the graphical login screen.

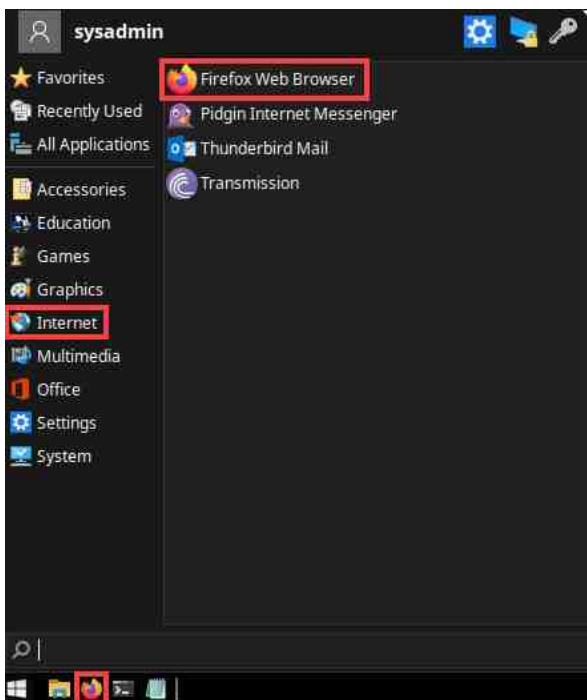


To launch the console window for a virtual machine, you may access by either clicking on the machine's graphic image from the topology page or by clicking on the machine's respective tab from the navigation bar.

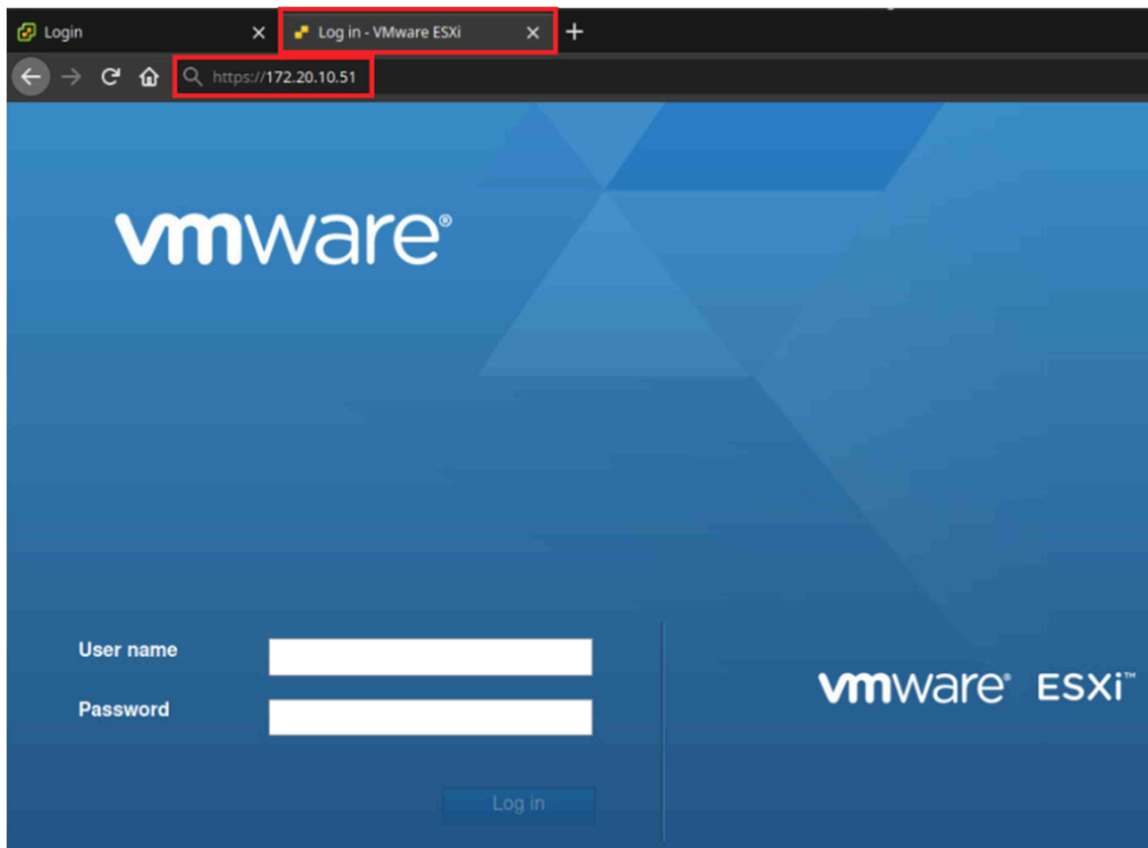
2. Log in as sysadmin using the password vmware123.



3. Launch the Mozilla Firefox web browser by either clicking on the icon shortcut found on the bottom toolbar or by navigating to Start Menu > Internet > Firefox Web Browser.

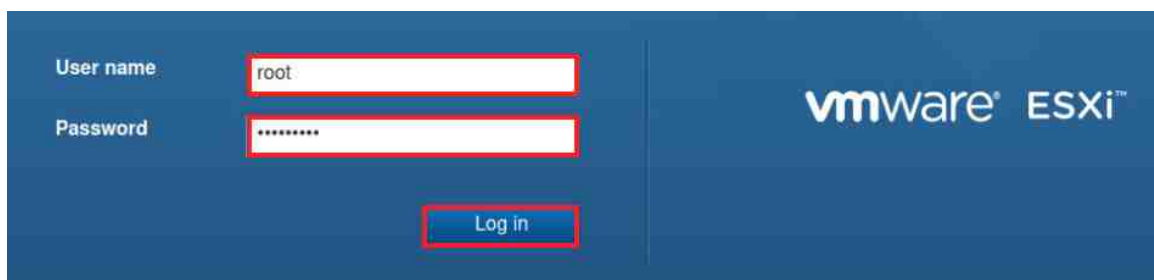


4. Notice the homepage is automatically configured to load the URL address of the VMware vCenter Server Appliance (sa-vcsa virtual machine). For this lab, you will open a new tab and type `https://172.20.10.51` in the address bar and click Enter. You will use sa-esxi-01 to complete this lab.

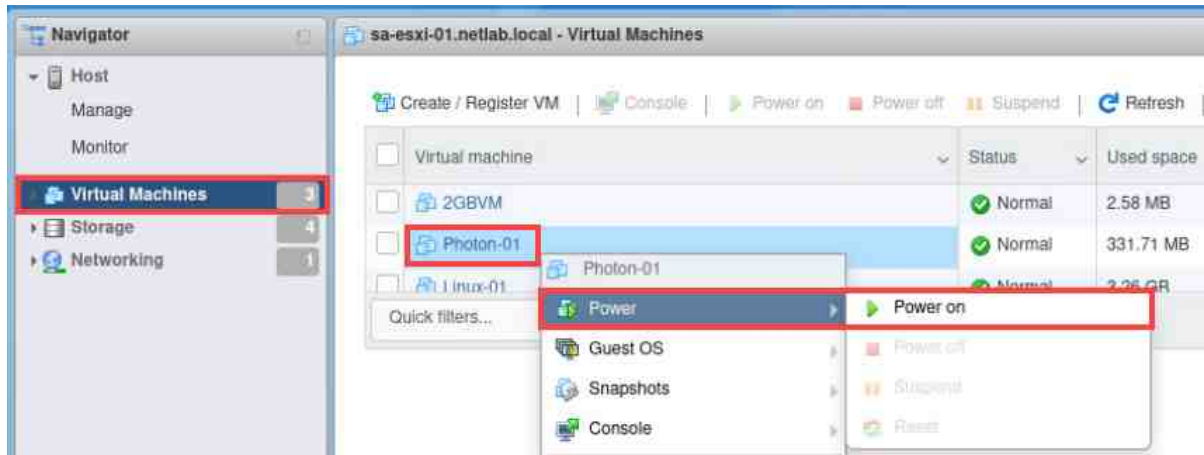


If you cannot successfully load the VMware ESXi webpage, then please wait an additional 3-5 minutes and refresh the page to continue. During this time, the ESXi host is still booting up and requires extra time to initialize.

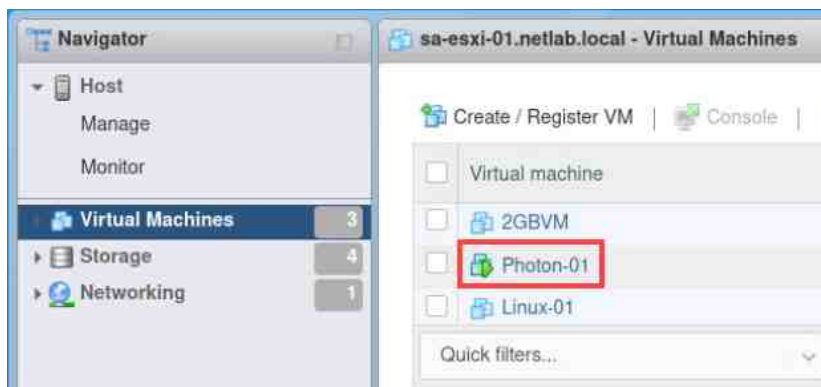
5. In the Username text field, type `root`, and in the Password text field, type `Training$`. Click on Log in.



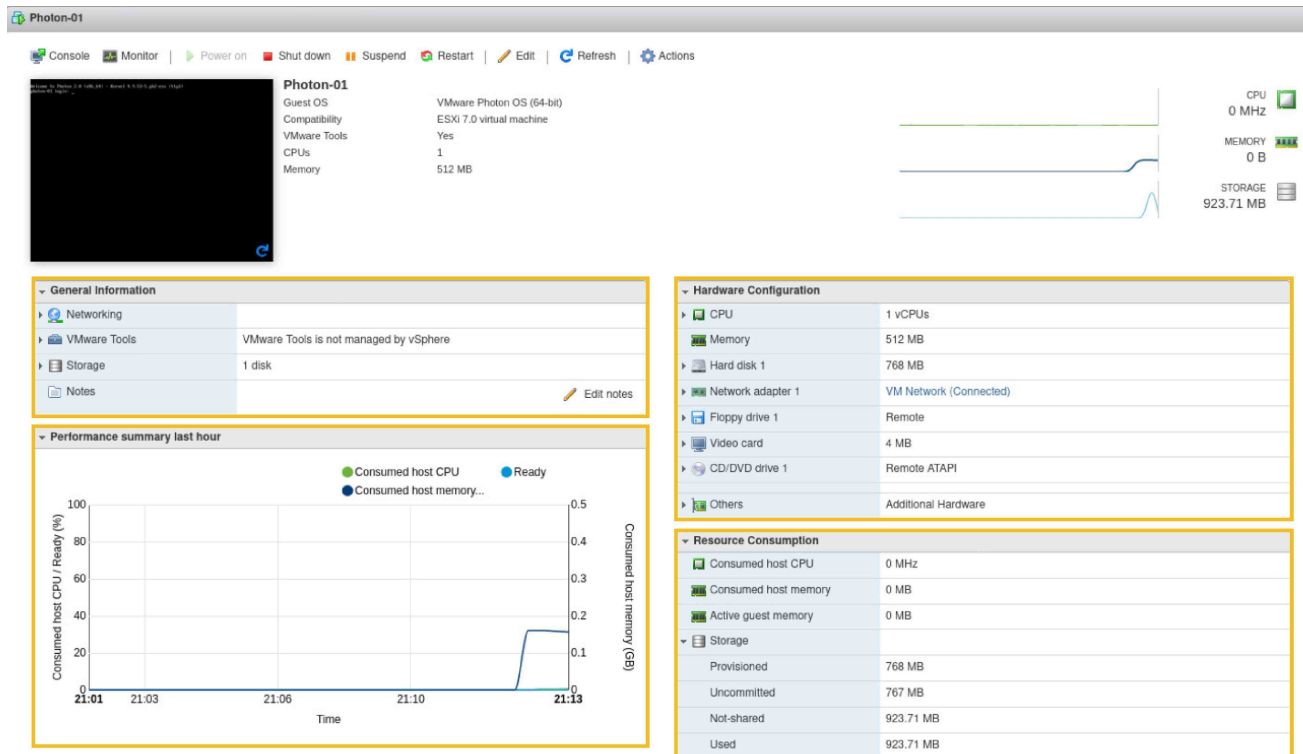
6. In the navigator pane, Click on the Virtual Machines object. In the sa-esxi-01.netlab.local – Virtual Machines window, right-click Photon-01, and Power > Power on.



7. In the right pane, click on the Photon-01 virtual machine.



8. Explore the General Information, Hardware Configuration, Performance summary last hour, and Resource Consumption windows.



9. Minimize the Recent tasks pane by clicking the Minimize icon at the top-right of the Recent tasks pane in the VMware Host Client.

Recent tasks						
Task	Target	Initiator	Queued	Started	Result	Completed
Power Off VM	Photon-01	root	07/22/2020 16:06:55	07/22/2020 16:06:55	Completed successfully	07/22/2020 16:06:55
Power On VM	Photon-01	root	07/22/2020 16:25:57	07/22/2020 16:25:57	Completed successfully	07/22/2020 16:25:57

10. View the Hardware Configuration pane, notice it contains the virtual hardware needed for the Photon-01 virtual machine to operate.

Hardware Configuration	
CPU	1 vCPUs
Memory	512 MB
Hard disk 1	768 MB
Network adapter 1	VM Network (Connected)
Floppy drive 1	Remote
Video card	4 MB
CD/DVD drive 1	Remote ATAPI
Others	Additional Hardware

11. Expand Hard disk 1 to view the hard disk information regarding the Backing, Capacity, Thin provisioned, Controller and Mode.

Hard disk 1	
Backing	[datastore 1] Photon-01/Photon-01-000001.vmdk
Capacity	768 MB
Thin provisioned	Yes
Controller	SCSI controller 0:0
Mode	Dependent



The purpose of the Hard disk 1 window is to provide a quick overview of the virtual hard disk configurations. The Backing shows what datastore the virtual machine is stored on. The Capacity shows the size of the Hard disk. Thin provisioned lets you know if the disk is thin or thick provisioned. Controller shows the version of storage controller that is needed to access the virtual disk. Mode lets you know the behavior of the VMDK file when a snapshot is taken.

12. View the Resource Consumption pane. Notice it indicates the resource consumption the Photon-01 virtual machine is using.

▼ Resource Consumption	
 Consumed host CPU	11 MHz
 Consumed host memory	187 MB
 Active guest memory	5 MB
▼ Storage	
Provisioned	768 MB
Uncommitted	745 MB
Not-shared	946.03 MB
Used	946.03 MB



The purpose of the Resource Consumption window is to show how much datastore capacity is being utilized by virtual machine files. These files can consist of configuration files, log files, and snapshots.

13. View the General Information window. Notice it contains specific information related to Photon-01. Leave VMware Host Client open and continue to the next task.

▼ General Information	
▼ Networking	
Host name	photon-01
IP addresses	1. 172.20.10.203 2. fe80::250:56ff:feaf:395f
▶ VMware Tools	VMware Tools is not managed by vSphere
▶ Storage	1 disk
Notes	 Edit notes

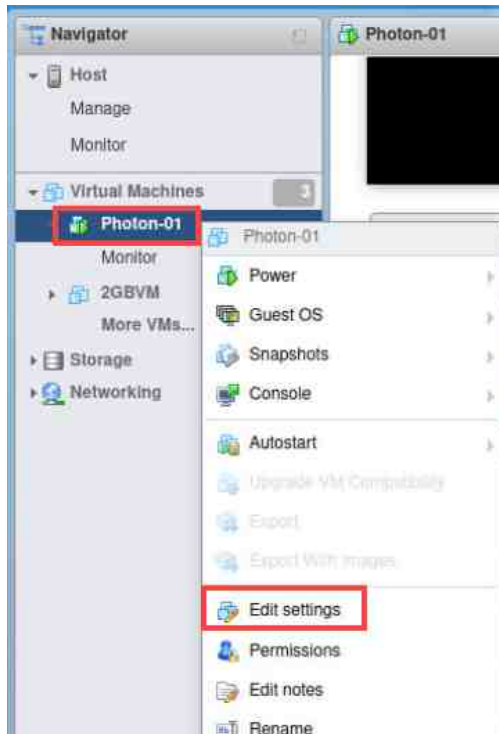


The purpose of the General Information window is to show specific configurations of a virtual machine. These configurations contain the Host name of the VM, IP address, and the status of VMware Tools. The Storage will show you how many active disks are associated and currently being utilized by the virtual machine.

2 Create and add Virtual Hard Disks to a Virtual Machine

In this task, you will go through the process of adding two virtual hard disks. You will also configure the first hard disk as thin-provisioned and the second as thick-provisioned.

1. In the Navigator pane, right-click on Photon-01 and click Edit Settings.



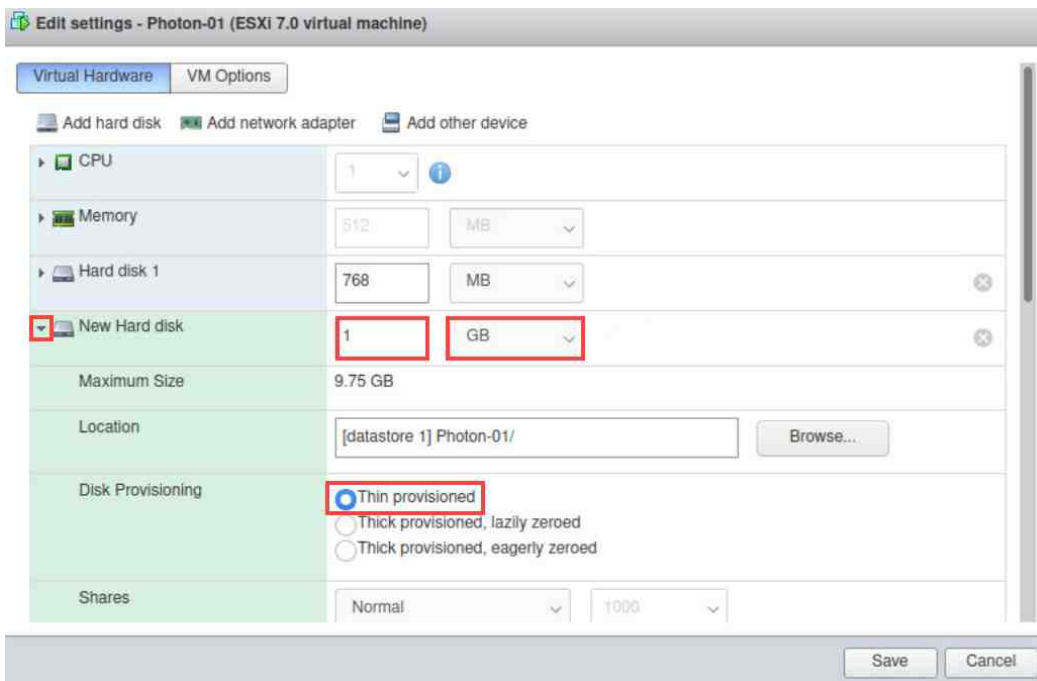
2. In the Edit settings window, Virtual Hardware tab, select Add hard disk and click New standard hard disk.



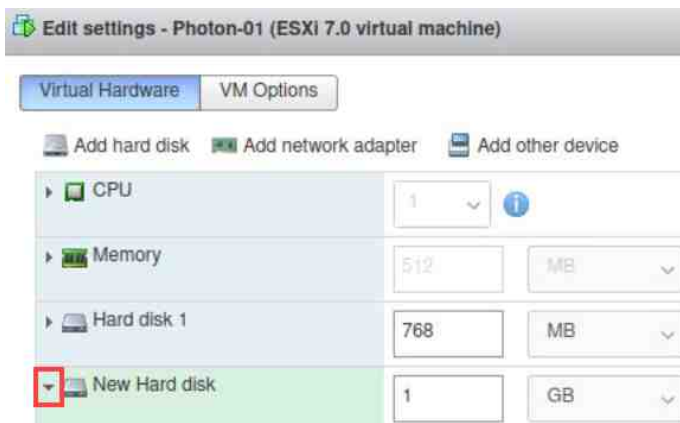
3. Notice the warning window that appears when adding a new hard disk. It is stating that you cannot overcommit disk space. You should increase the datastore capacity or reduce the size of the disk. You may close the warning window and continue to the next step.



4. In the Edit settings window, expand the New Hard disk and change the disk size to 1 GB. Select Thin provisioned for the Disk Provisioning.



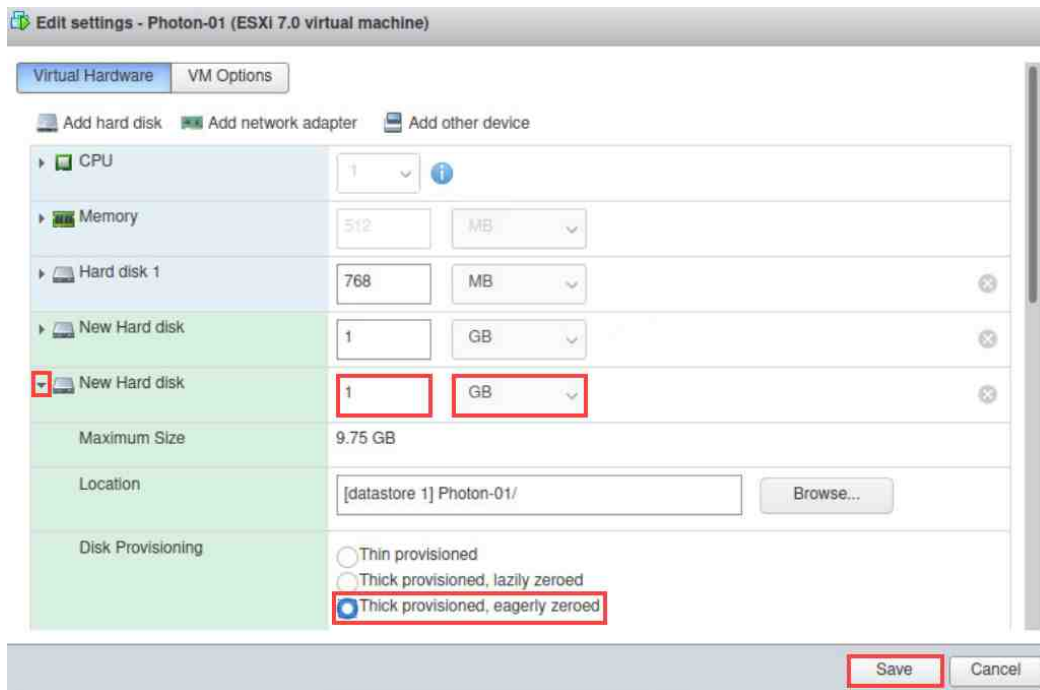
5. Minimize the New Hard Disk.



- While on the Virtual Hardware tab, add another hard disk by selecting Add hard disk and click New standard hard disk.



- In the Edit settings window, expand the New Hard disk and change the disk size to 1 GB. Select Thick provisioned, eagerly zeroed for the Disk Provisioning. Lastly, click Save.



8. In the Photon-01 main workspace window, in the Hardware Configuration pane, ensure Hard disk 2 and Hard disk 3 are expanded. Verify that Hard disk 2 is set to 1 GB and is a thin provisioned disk. Verify that Hard disk 3 is set to 1 GB and says No for Thin provisioned. You will confirm Hard disk 3 is a Thick Provisioned Disk in Task 3.

Hardware Configuration	
CPU	1 vCPUs
Memory	512 MB
Hard disk 1	768 MB
Hard disk 2	
Backing	[datastore 1] Photon-01/Photon-01_1.vmdk
Capacity	1 GB
Thin provisioned	Yes
Controller	SCSI controller 0:1
Mode	Dependent
Hard disk 3	
Backing	[datastore 1] Photon-01/Photon-01_2.vmdk
Capacity	1 GB
Thin provisioned	No
Controller	SCSI controller 0:2
Mode	Dependent
Network adapter 1	VM Network (Connected)

9. Leave the Photon-01 virtual machine selected on the ESXi Host Client and continue to the next task.

3 Differentiate between Thin-Provisioned and Thick-Provisioned Disk

In this task, you will view and compare a thin-provisioned and thick provisioned disk.

1. On the Hardware Configuration pane, ensure Hard disk 2 and Hard disk 3 are expanded.

Hardware Configuration	
CPU	1 vCPUs
Memory	512 MB
Hard disk 1	768 MB
Hard disk 2	
Backing	[datastore 1] Photon-01/Photon-01_1.vmdk
Capacity	1 GB
Thin provisioned	Yes
Controller	SCSI controller 0:1
Mode	Dependent
Hard disk 3	
Backing	[datastore 1] Photon-01/Photon-01_2.vmdk
Capacity	1 GB
Thin provisioned	No
Controller	SCSI controller 0:2
Mode	Dependent

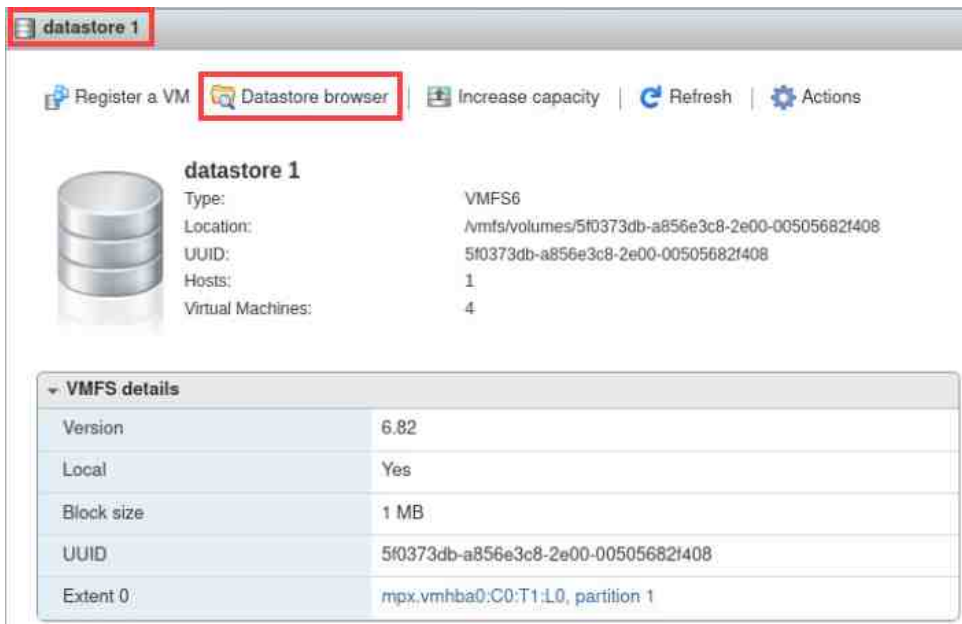
2. In the Hardware configuration pane, notice the name of the datastore on which Hard disk 2 and Hard disk 3 reside and the name of the virtual disk files.

Hardware Configuration	
CPU	1 vCPUs
Memory	512 MB
Hard disk 1	768 MB
Hard disk 2	
Backing	[datastore 1] Photon-01/Photon-01_1.vmdk
Capacity	1 GB
Thin provisioned	Yes
Controller	SCSI controller 0:1
Mode	Dependent
Hard disk 3	
Backing	[datastore 1] Photon-01/Photon-01_2.vmdk
Capacity	1 GB
Thin provisioned	No
Controller	SCSI controller 0:2
Mode	Dependent

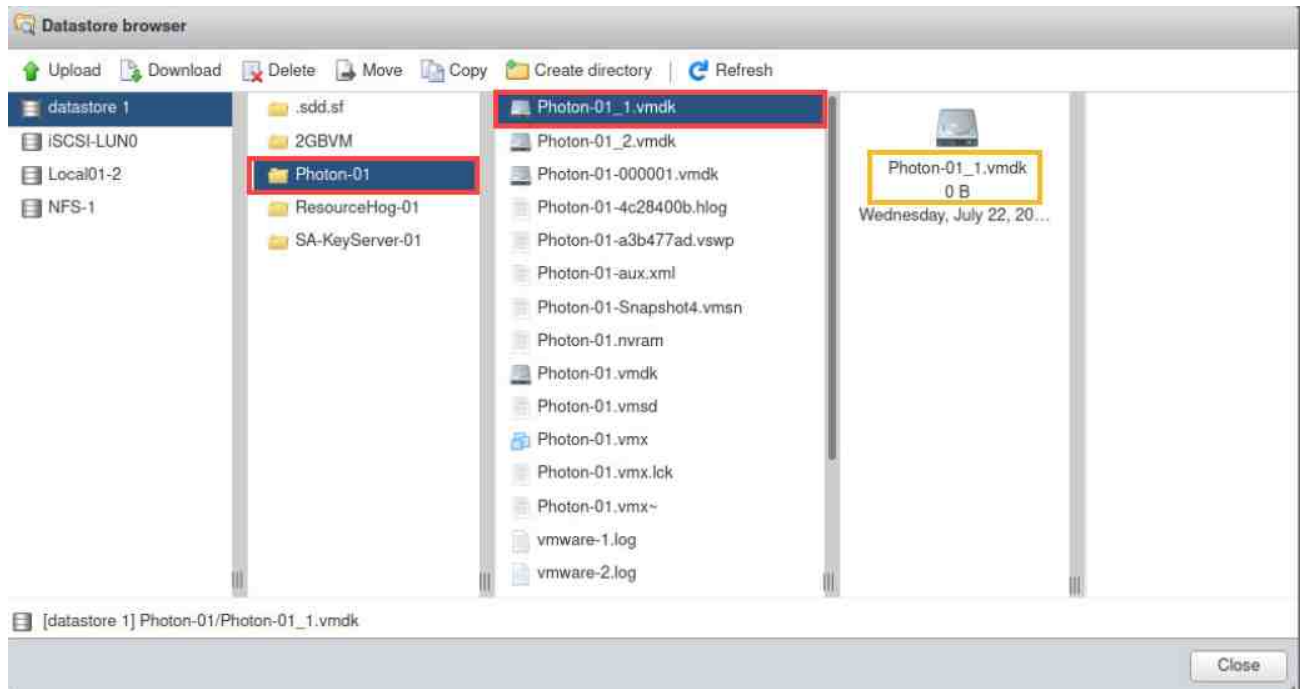
3. In the Navigator pane, select Storage and click datastore 1.



4. In the datastore 1 window, click Datastore browser.

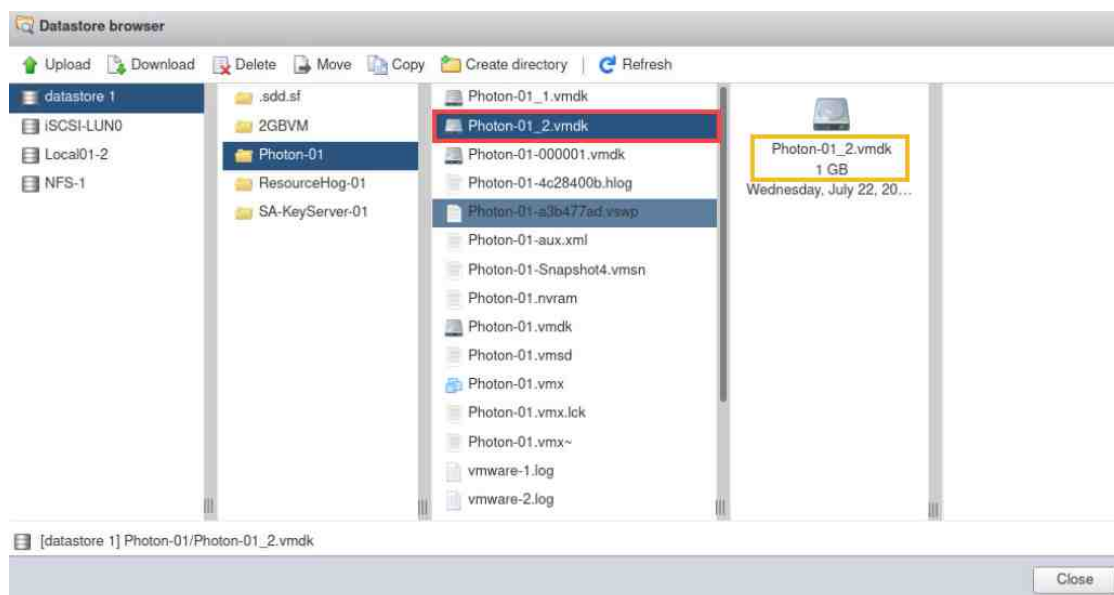


5. In the Datastore browser window, select the Photon-01 folder and select the Photon-01_1.vmdk.



A thin provisioned disk only utilizes as much datastore space as needed on the disk. Notice that the disk size is 0 bytes. It is 0 bytes due to no storage being utilized on the disk in its current state. If a thin provisioned disk needs more space, it will inflate up to its maximum capacity and eventually absorb the entire disk.

6. In the Datastore browser window, select the Photon-01_2.vmdk.





A thick provisioned disk utilizes the entire disk during creation. Notice that the size of the disk is 1 GB. During creation of the virtual storage disk you allocated 1 GB of space therefore it gets all the space allocated to it at the time of creation.

7. Close the Datastore browser window. Leave the VMware Host Client open and continue to the next task.

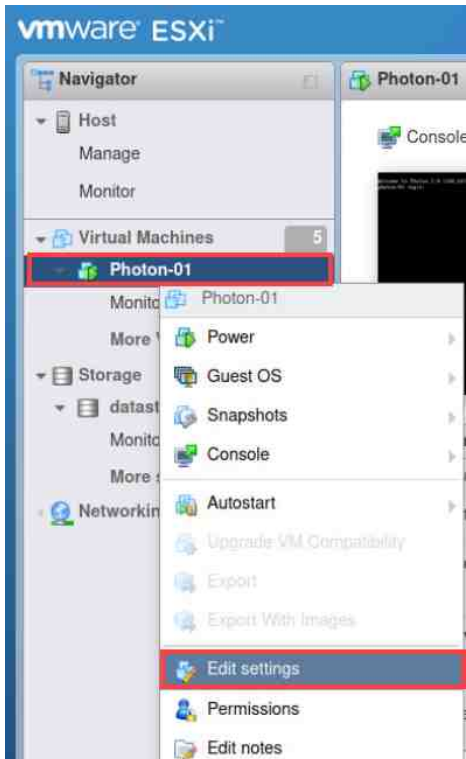
4 Delete Virtual Hardware

In this task, you will delete the virtual hardware you created in a previous task.

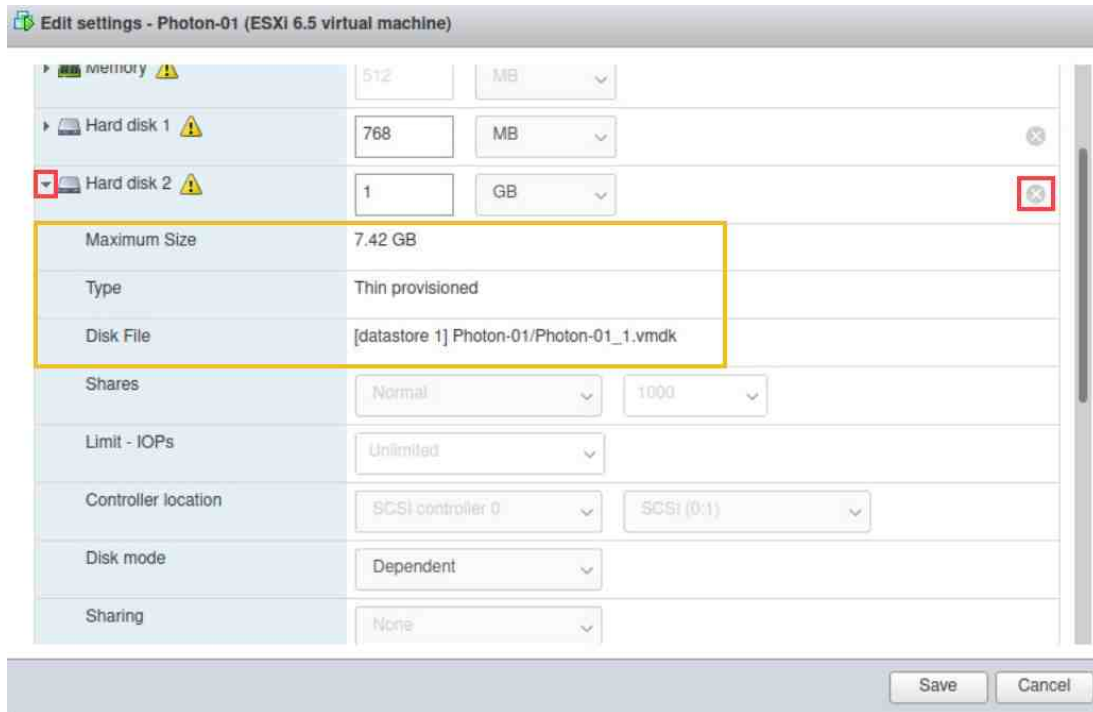
1. While in the VMware Host Client, in the Navigator pane, select the Photon01 virtual machine.



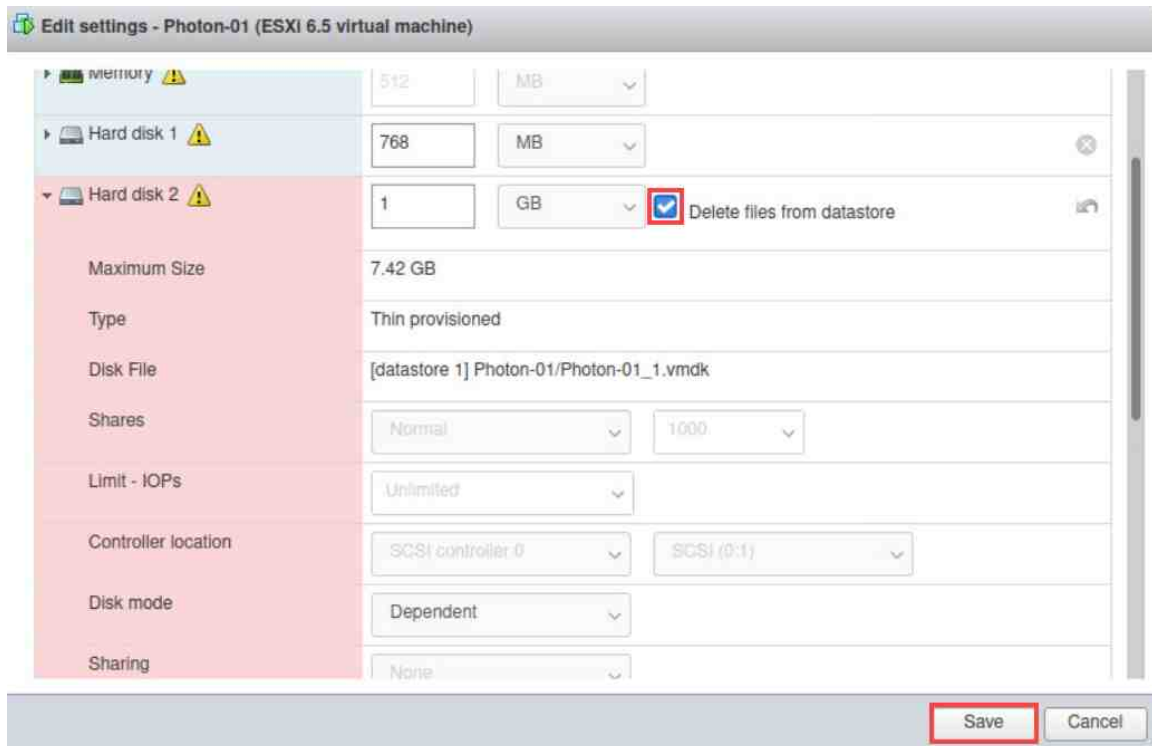
2. Right-click on the Photon-01 virtual machine and click Edit Settings.



- On the Edit settings window, expand Hard disk 2. Notice that Hard disk 2 is Thin provisioned and the file name of Photon-01_1.vmdk. Click the x icon located to the far right to delete Hard disk 2.



- Select Delete files from datastore and click Save.



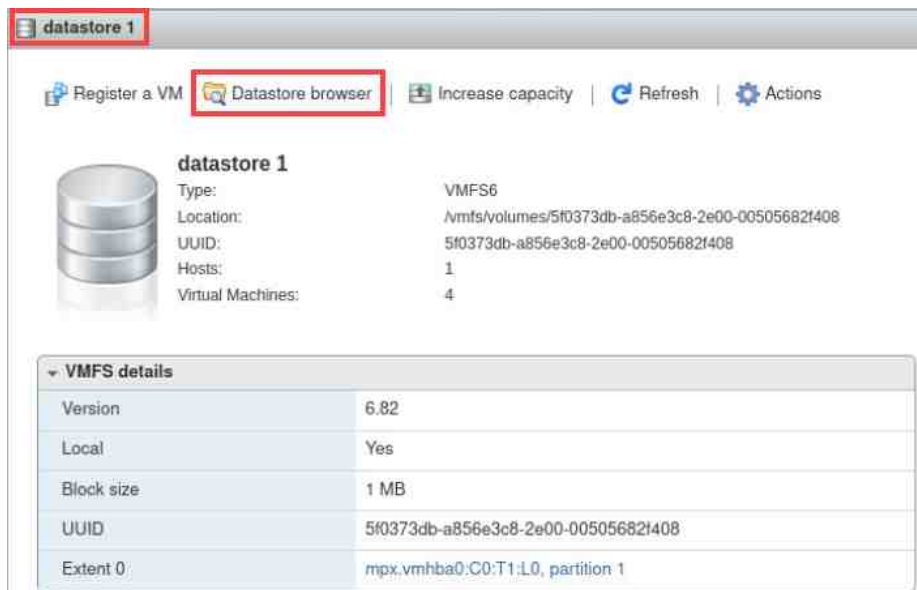
Please Note

Once you click Save, this will delete Hard disk 2 from the Photon-01 virtual machine. Having selected to Delete files from datastore, this will remove all the disk files associated with Hard disk 2 from datastore 1.

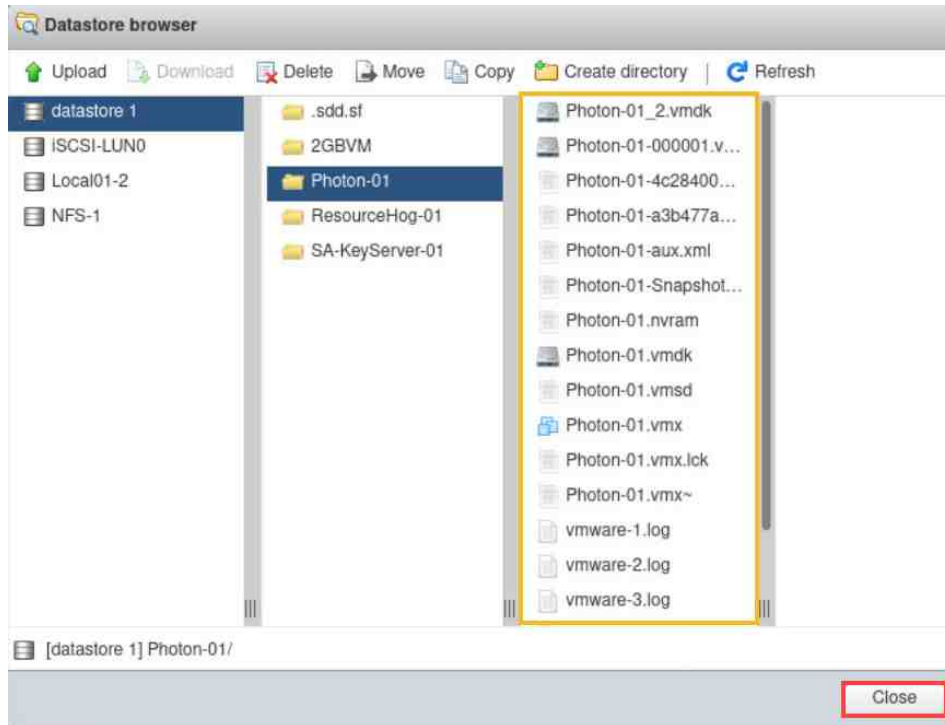
5. Change focus back to the Navigator pane, expand Storage, and click datastore 1.



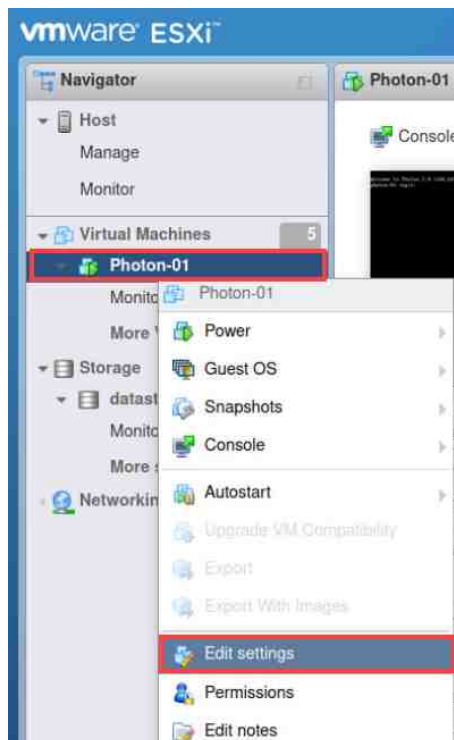
6. In the datastore 1 window, click Datastore browser.



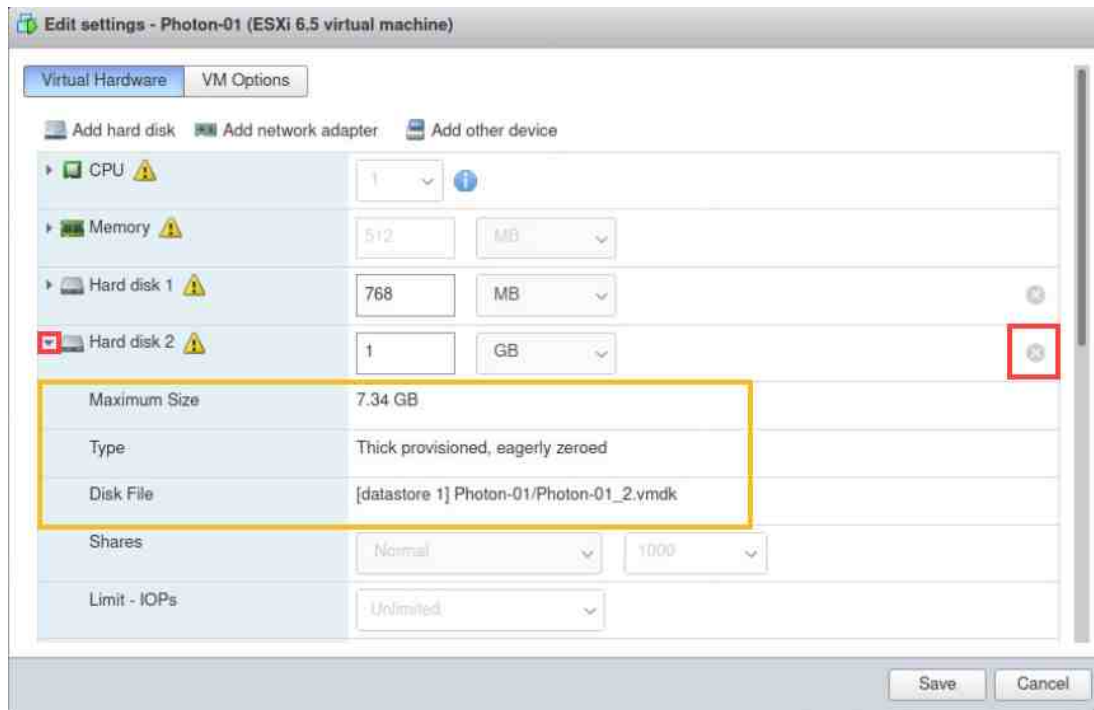
7. In the Datastore browser window, select the Photon-01 folder. Notice the disk files Photon-01_1.vmdk have been deleted. The files associated with Hard disk 2 were deleted due to selecting Delete files from datastore. Click Close.



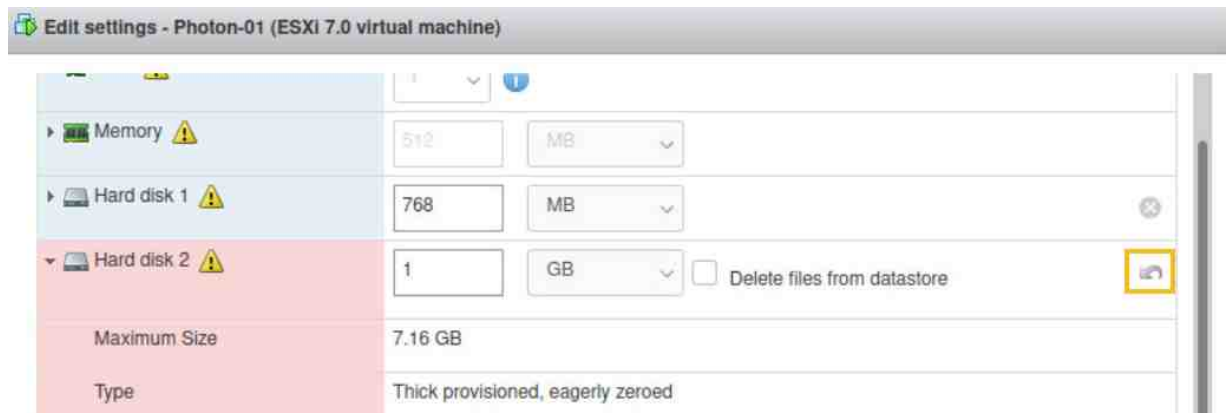
8. In the Navigator pane, right-click Photon-01 and click Edit Settings.



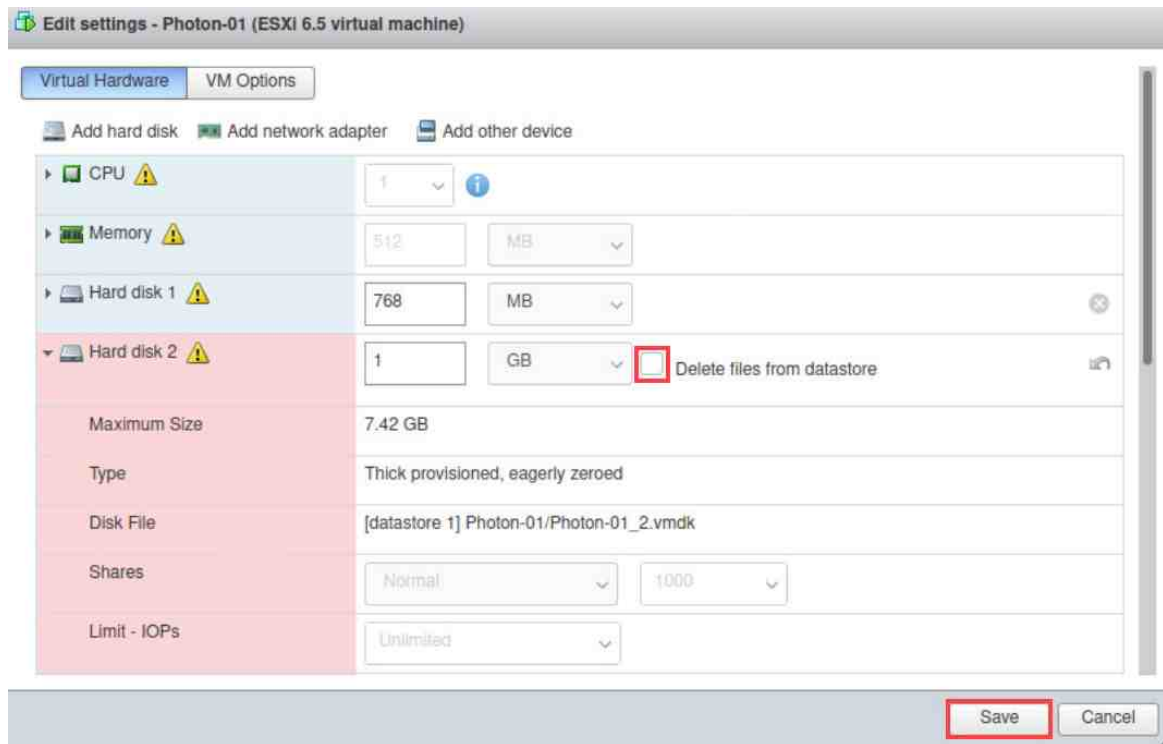
9. On the Edit settings window, expand Hard disk 2. Notice Hard disk 3 that was created in a previous task has been renamed to Hard disk 2. Since you deleted the original Hard disk 2 from the Photon-01 virtual machine and datastore 1, it now only sees two disks to be utilized. Verify that the newly renamed disk is the Thick provisioned, eagerly zeroed disk with a file name of Photon-01_2.vmdk that was created in a previous task. Click the x icon located to the right of the newly named Hard disk 2.



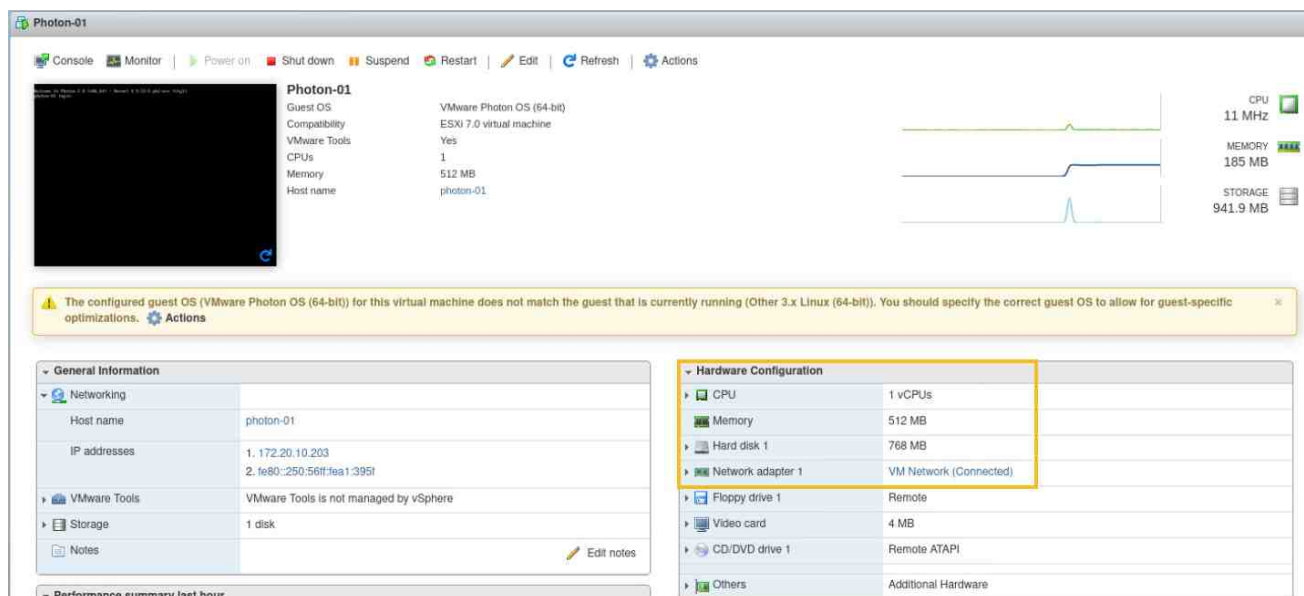
10. On the Edit settings window, notice the reverse icon located at the far-right. The reverse icon is there to keep you from deleting a hard disk or other virtual hardware without having to close the window and lose other configurations.



11. Verify Delete files from datastore is not selected and click Save.



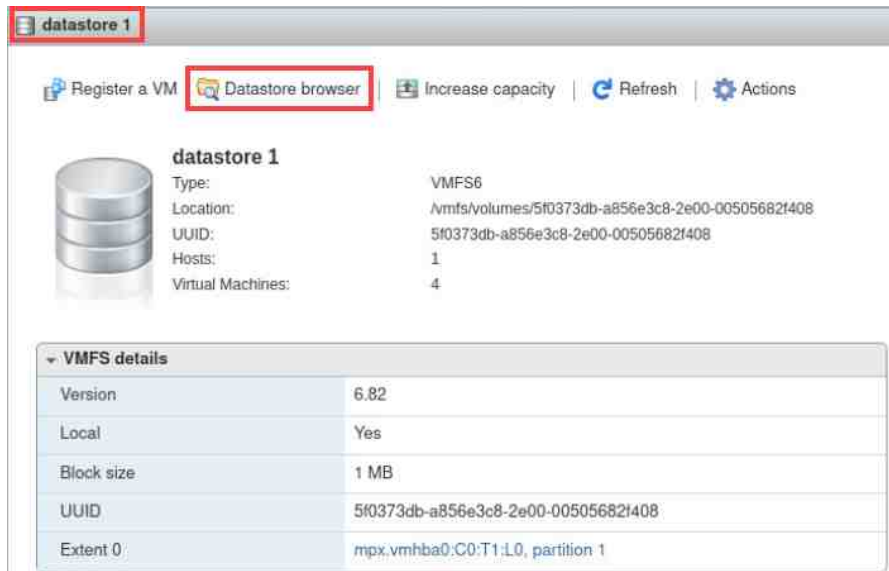
12. In the Hardware configuration pane of Photon-01, verify that Hard disk 2 has been deleted from Photon-01.



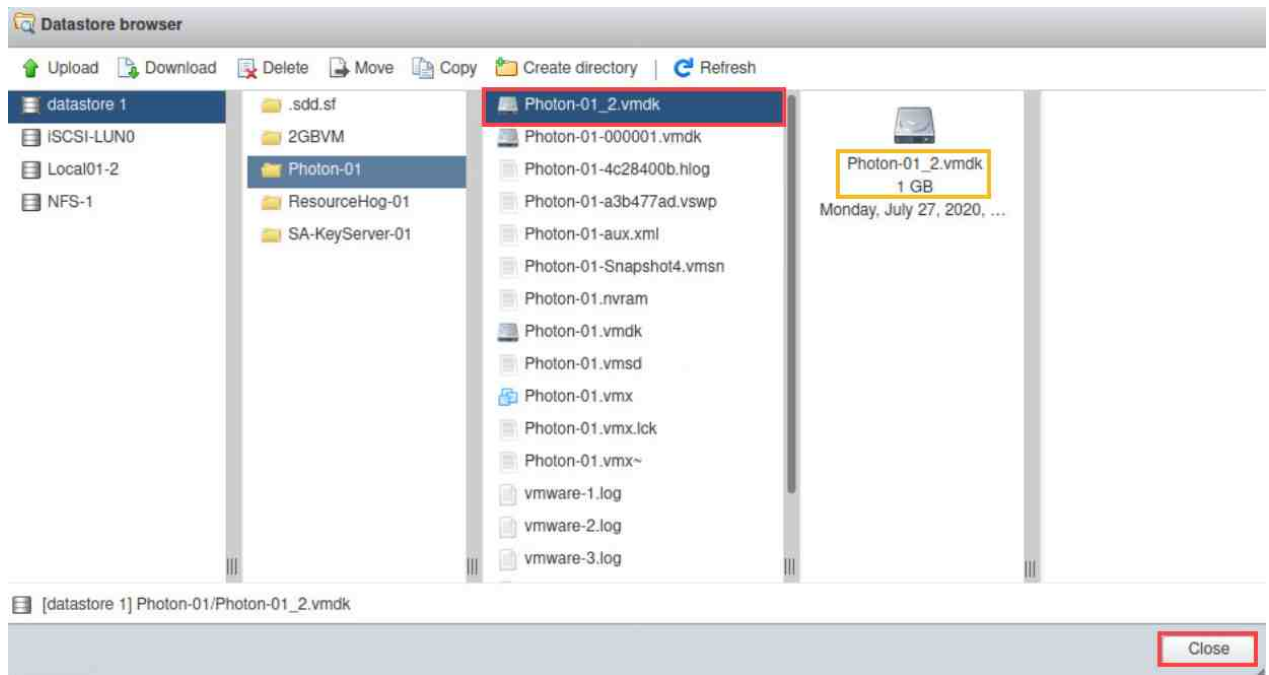
13. Change focus back to the Navigator pane, expand Storage, and click datastore 1.



14. In the datastore 1 window, click Datastore browser.



15. In the Datastore browser window, select the Photon-01 folder. Notice the Photon-01_2.vmdk is still present. The disk files associated with Hard disk 2 were not deleted. Hard disk 2 has been deleted, but the files will remain in Datastore 1 until they are deleted permanently from the disk. Click Close.



16. The lab is now complete; you may end the reservation.